

Frequency-Domain ILC design

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Huge progress in learning

- ▶ (video) games: backgammon, chess, go, Atari, starcraft, ...
- ▶ LLMs
- ▶ ...

Enabled by

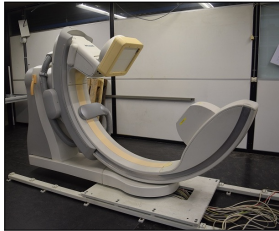
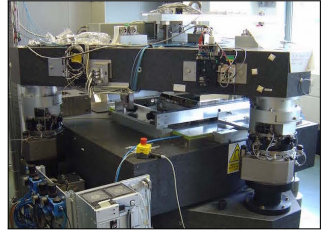
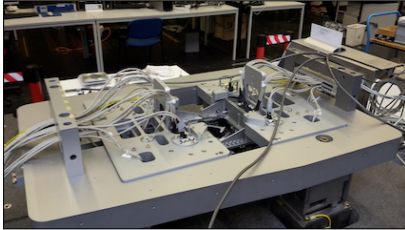
- ▶ big data
- ▶ ubiquitous computing (e.g., cloud services)



What does learning have to offer for real-world systems?

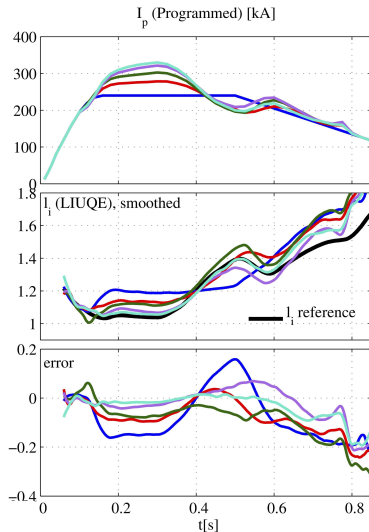
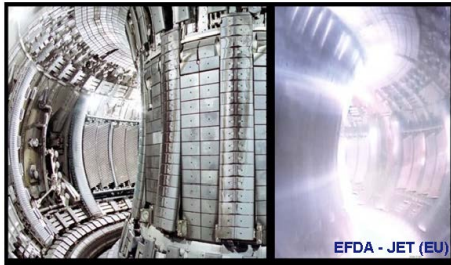
Learning in Machines

- ▶ repeating tasks
- ▶ experiments inexpensive
- ▶ but still (Oomen2018b)
 - ▶ real-time
 - ▶ no parallelization
 - ▶ time variation
wear, aging, ...

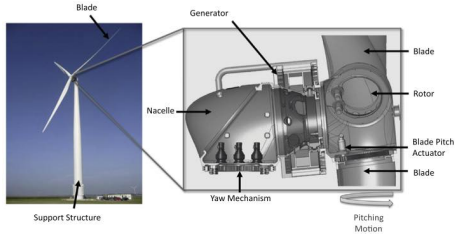


Nuclear fusion: TCV Tokamak, Lausanne^(Felici & Oomen2015)

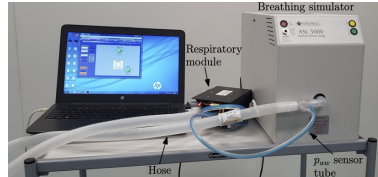
- ▶ again repeating tasks
- ▶ for successful operation internal inductance I_i must follow a reference
- ▶ by adjusting total plasma current I_p
- ▶ experiments are rather expensive



Also energy systems (Navalkar et al.2014)



and mechanical ventilation (Reinders et al.2020)



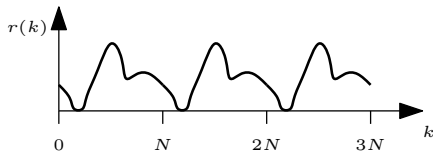
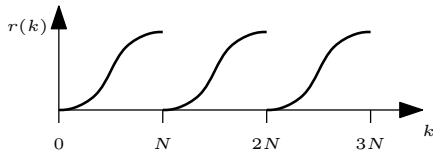
Very similar: repeating tasks in continuous operation (no reset)

Key properties

- ▶ fast: merge model-based approaches and data-driven learning
- ▶ safe: provide robustness for safe operation

Main techniques

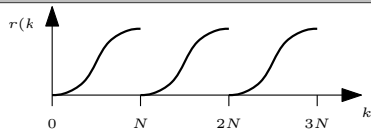
- ▶ Iterative Learning Control (ILC): batch operation
- ▶ Repetitive Control (RC): continuous operation



Successfully applied in all previous slides

Iterative learning control

- ▶ initial ideas
 - ▶ 1967 US patent 3555252: "Learning control of actuators in control systems"
 - ▶ 1984 paper (Arimoto et al.1984): "Bettering operation of robots by learning"
- ▶ 1990-present: large number of design frameworks and successful applications
 - ▶ frequency-domain designs (Bristow et al.2006)(Moore1999)(Moore1993)
 - ▶ norm-optimal (Gunnarsson & Norrlöf2001)(Owens2016)
 - ▶ nD systems theory (Rogers et al.2007)
 - ▶ nonlinear (Bien & Xu1998)

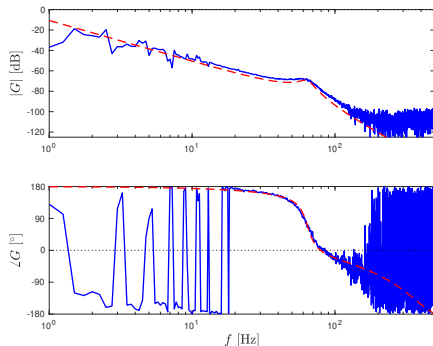


Today: frequency domain design

- ▶ useful if frequency response functions are common
 - ▶ e.g., mechatronics!

Repetitive Control

- ▶ initial contributions (Francis & Wonham1976) (Inoue et al.1981)
- ▶ 1990s-present: very similar design (Blanken et al.2020)



Iterative Learning Control - What Can Be Achieved?

Traditional Motion Control

ILC: Basic Principles

Convergence Analysis

Enforcing Robustness

Implementation Aspects

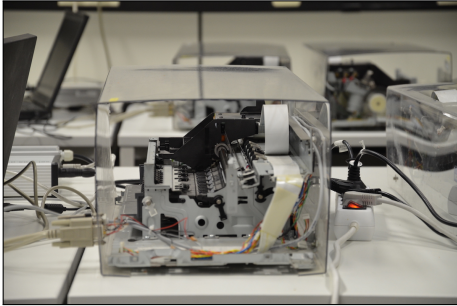
Iteration-Variant Disturbances*

Alternative setup: serial ILC*

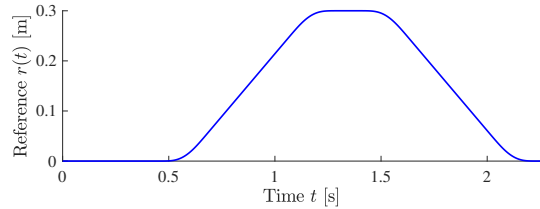
Multivariable ILC*

Summary

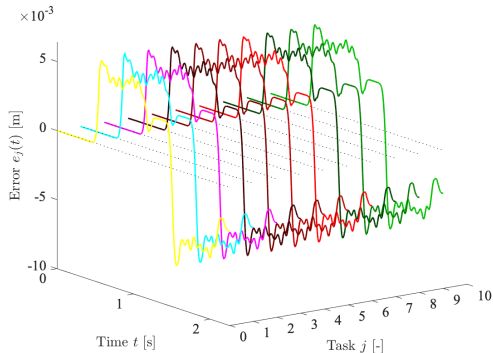
Consumer printer system example



Reference



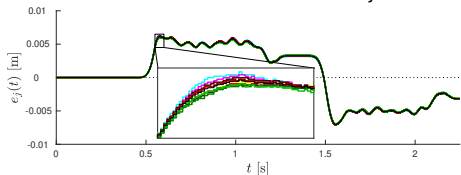
Traditional motion control^(Oomen2020a)



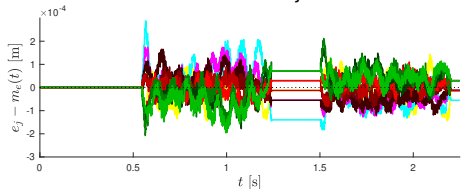
Highly reproducible error $e_j(t)$

Procedure^(Oomen2020b)

Step 1: compute $m_e(t) = \frac{1}{n_{\text{exp}}} \sum_{j=0}^{n_{\text{exp}}-1} e_j(t)$:

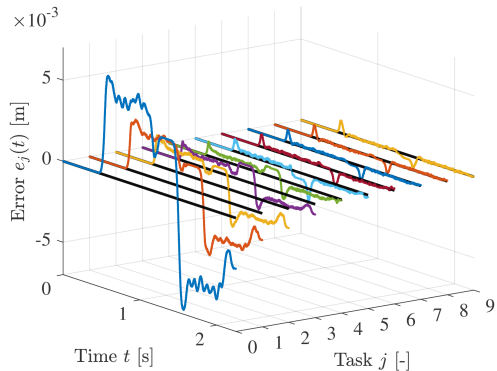


Step 2: ILC can achieve $e_j(t) - m_e(t)$:



How to achieve this?

Spoiler



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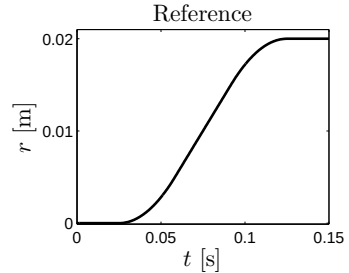
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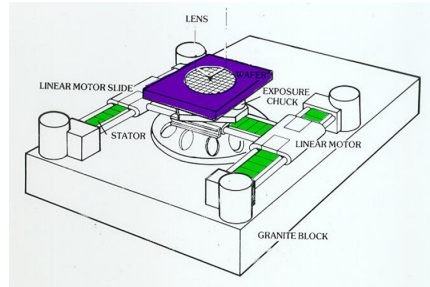
Servo control

- ▶ manipulate input such that output is close to reference
- ▶ e.g., motion systems



Control

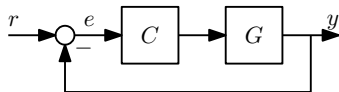
- ▶ feedback
- ▶ feedforward



Feedback

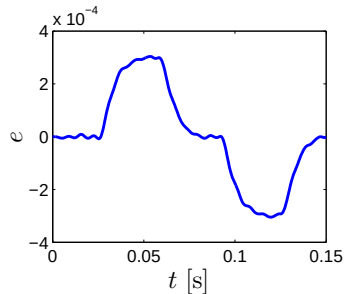
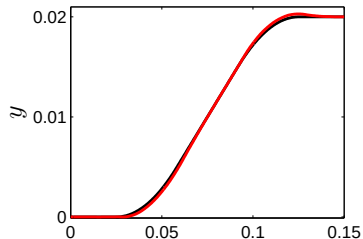
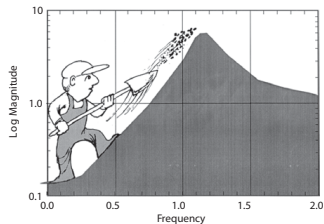
► deal with uncertainty

- disturbances
- system dynamics



► performance limits

- Bode sensitivity integral: **causality**



Feedforward

- ▶ compensate for *known* disturbances
 - ▶ e.g., reference
- ▶ Q: how to choose f ?
- ▶ A: note that

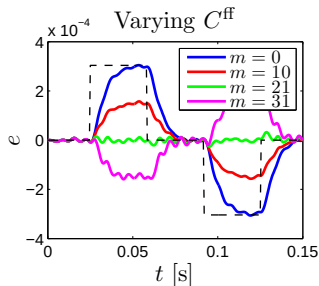
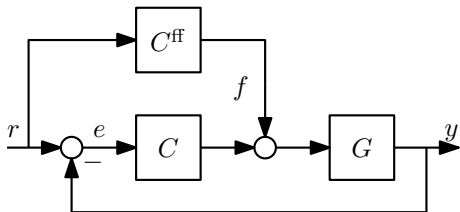
$$e = r - Gf - GCe$$

$$(1 + GC)e = r - Gf$$

$$e = \underbrace{(1 + GC)^{-1}}_{\text{feedback}} \underbrace{(r - Gf)}_{\text{feedforward}},$$

Pick $f = C^{\text{ff}} r$ with $C^{\text{ff}} = G^{-1}$. Then, $e = 0$

- ▶ need to know r
- ▶ performance depends on 'model quality' G^{-1}



Applying the procedure of Slide 9

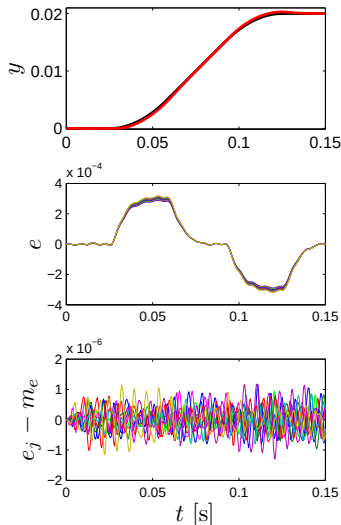
- ▶ 10 experiments without feedforward
- ▶ step 1: reproducible error

$$m_e(t) = \frac{1}{n_{\text{exp}}} \sum_{j=0}^{n_{\text{exp}}-1} e_j(t)$$

- ▶ step 2: non-repeating error
 - ▶ $e_j - m_e, j = 1, \dots, 10$
 - ▶ small!

Next: ILC

- ▶ compensate the unknown 'trial-invariant' disturbance that induces m_e



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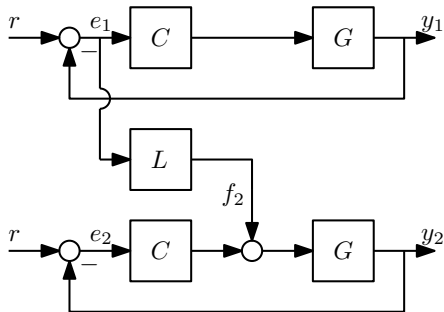
Implementation Aspects

Iteration-Variant Disturbances*

Alternative setup: serial ILC*

Multivariable ILC*

Summary



Experiment 1

- ▶ measure e_1
- ▶ assume r not measured

Experiment 2

- ▶ identical r
- ▶ which f_2 for small e_2 ?

linearity:

$$e_2 = e_1 - GSf_2$$

$$\text{let } f_2 = Le_1$$

$$\Rightarrow e_2 = e_1 - GSLe_1$$

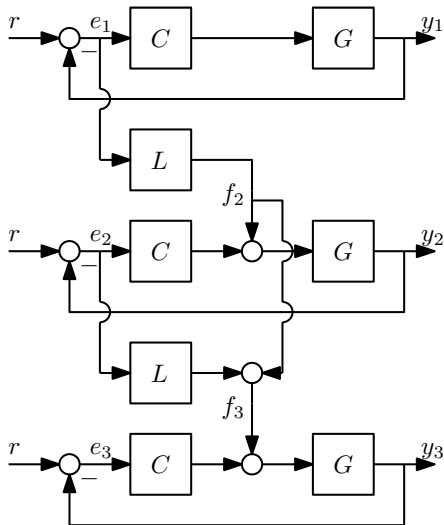
$$\text{then } L = (GS)^{-1}$$

$$\Rightarrow e_2 = 0$$

linearity:

$$e_2 = e_1 - GSf_2$$

$$\text{let } f_2 = Le_1$$



Experiment 2

► $e_2 = e_1 - G S L e_1$

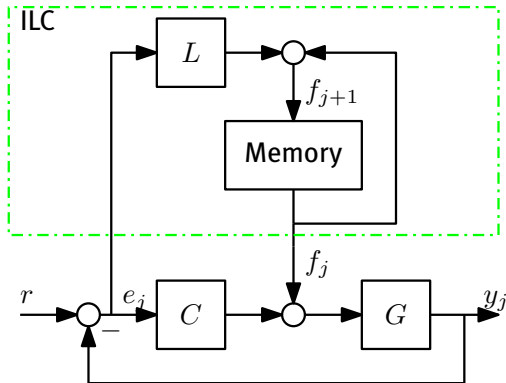
Model errors

- in general: L^{-1} not exact
 $\Rightarrow e_2 \neq 0$

Idea

- do the same trick again!
- $f_{j+1} = \underbrace{f_j}_{\text{previous feedforward}} + \underbrace{L e_j}_{\text{correction}}$
- **iterative** learning!

Basic ILC scheme

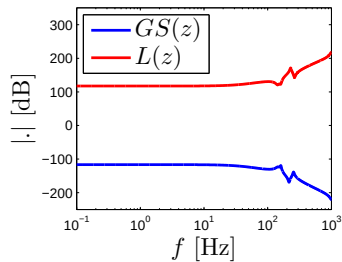
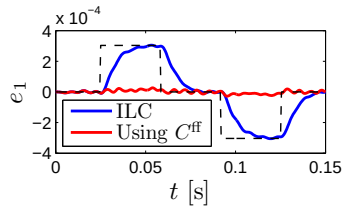
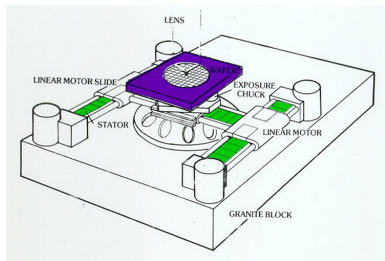


ILC algorithm

- ▶ learning update:
 - ▶ $f_{j+1} = f_j + Le_j$
 - ▶ compute off-line
- ▶ discrete time implementation:
 - ▶ memory: vector of data
 - ▶ ILC using
 - ▶ discrete time signals
 - ▶ discrete time systems

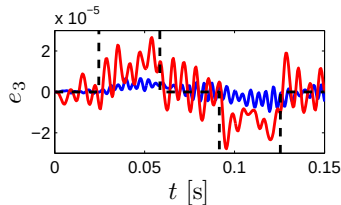
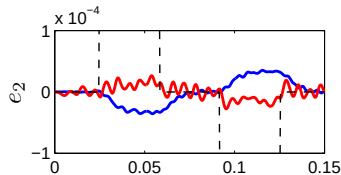
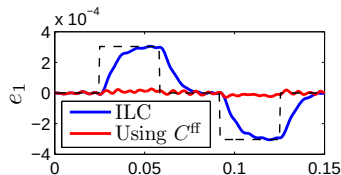
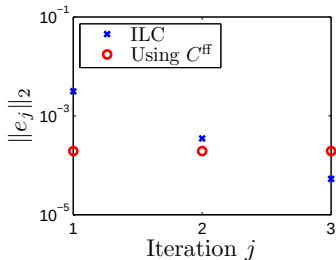
Next

- ▶ example



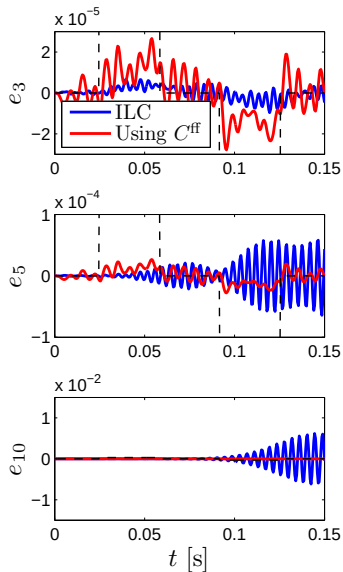
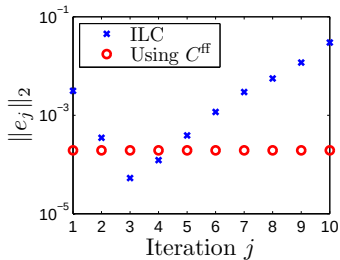
ILC iterations

1. Initial error
2. After one iteration:
 - ▶ error reduced
3. After two iterations:
 - ▶ error further reduced
 - ▶ outperforms C^{ff} !
 - ▶ more iterations?



More ILC iterations

- ▶ **increasing** error in experiment 4, 5
- ▶ error increases for $j = 6, 7, \dots, 10$
- ▶ what's wrong?



Iterative Learning Control - What Can Be Achieved?

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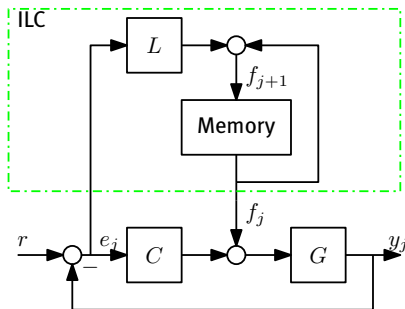
Iteration-Variant Disturbances*

Alternative setup: serial ILC*

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Summary

Basic ILC scheme



ILC algorithm

- ▶ learning update:

$$f_{j+1} = f_j + L e_j$$

- ▶ $e_j = S r - G S f_j$
 $\Rightarrow f_j = (G S)^{-1} (S r - e_j)$

- ▶ Error propagation:

$$\begin{aligned} e_{j+1} &= S r - G S f_{j+1} \\ &= S r - G S (f_j + L e_j) \\ &= (1 - G S L) e_j \end{aligned}$$

- ▶ Does the iterative scheme converge?

ILC iteration

$$\blacktriangleright e_{j+1} = \underbrace{(1 - GSL)}_{=:F} e_j$$

Approach

- ▶ infinite time $\Rightarrow$ frequency domain conditions
- ▶ based on contraction mapping
- ▶ underlying result:

If F is a contraction, then the iteration $F(\dots (Fe_j) \dots)$ converges to a **unique** fixed point

Theorem Let $f : X \mapsto X$, with $d(\cdot, \cdot)$ a metric on X . Then, f is called a contraction if there exists a $k \in [0, 1)$ such that

$$d(f(x_1), f(x_2)) \leq k d(x_1, x_2), \quad \forall x_1, x_2 \in X$$

Interpretation

- ▶ points $f(x_1)$ and $f(x_2)$ closer together than x_1 and x_2
- ▶ take as x_2 as $e_\infty = f(e_\infty)$ and x_1 as e_j . Then e_j converges to e_∞

Application to ILC

- ▶ use norms to define appropriate metric
- ▶ ℓ_2 : square summable sequence, i.e., $\|x\|_2^2 = \sum_{i=-\infty}^{\infty} |x_i|^2 < \infty$
- ▶ then, $\|Fx_1 - Fx_2\|_2 = \|F(x_1 - x_2)\|_2 \leq \|F\|_{2,2} \|x_1 - x_2\|_2$

Key result: F is a contraction if $\|F\|_{2,2} < 1$

Induced ℓ_2 -norm

$$\blacktriangleright \|F\|_{2,2} = \|F\|_{\mathcal{L}_\infty} = \sup_{\omega \in [0, 2\pi)} |F(e^{j\omega})|$$

ILC convergence result

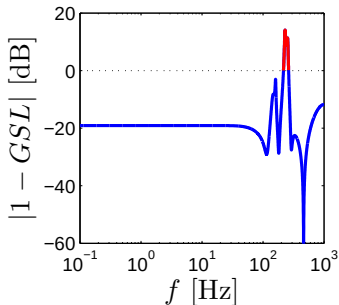
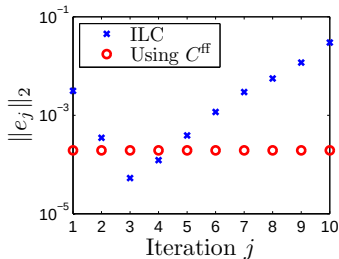
The ILC iteration

$$e_{j+1} = (1 - GSL)e_j$$

converges monotonically in the ℓ_2 norm if $|1 - G(e^{j\omega})S(e^{j\omega})L(e^{j\omega})| < 1 \quad \forall \omega \in [0, 2\pi)$

- $\blacktriangleright$ L often **noncausal**: condition based on $\mathcal{L}_\infty$ norm, and not $\mathcal{H}_\infty$ norm^(Oomen & Rojas2017)
- $\blacktriangleright$ can use a **measured** FRF of $G(e^{j\omega})S(e^{j\omega})$

Servo example again



ILC convergence

- ▶ sampling time $h = 5 \cdot 10^{-4}$
- ▶ monotonic convergence if $|1 - G(e^{j\omega h})S(e^{j\omega h})L(e^{j\omega h})| < 1, \quad \forall \omega \in [0, 2\pi \cdot 2000]$
- ▶ condition **violated** in example

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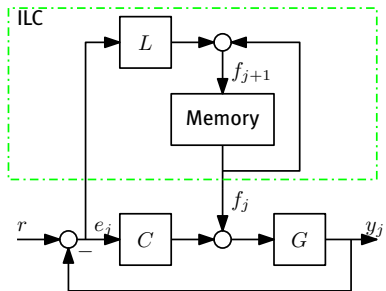
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- ▶ learning update:

$$f_{j+1} = f_j + L e_j$$

- ▶ error propagation:

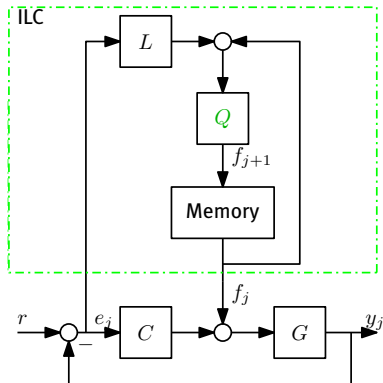
$$e_{j+1} = (1 - GSL) e_j$$

- ▶ monotonic convergence:

$$|1 - GSL| < 1, \quad \forall \omega \in [0, 2\pi) \quad (1)$$

- ▶ what if (1) is violated?

Basic ILC scheme



ILC algorithm

- ▶ introduce Q
- ▶ learning update:

$$f_{j+1} = Q(f_j + Le_j)$$

- ▶ error propagation:

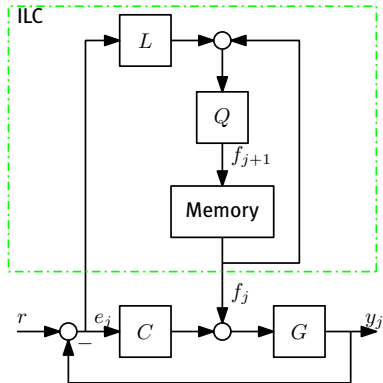
$$e_{j+1} = Q(1 - GSL)e_j + (1 - Q)Sr$$

- ▶ monotonic convergence:

$$|Q(1 - GSL)| < 1, \quad \forall \omega \in [0, 2\pi) \quad (2)$$

- ▶ exploit freedom in Q to satisfy (2)!

ILC scheme



ILC design procedure

- learning update:

$$f_{j+1} = Q(f_j + Le_j)$$

1. design L to approximate $(GS)^{-1}$

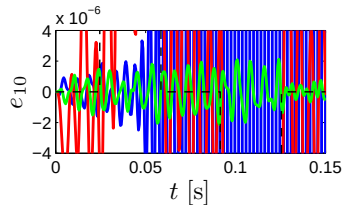
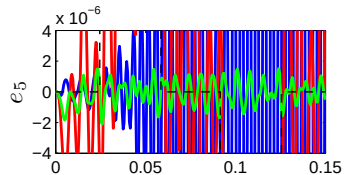
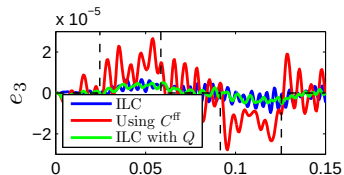
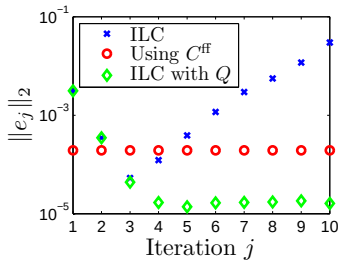
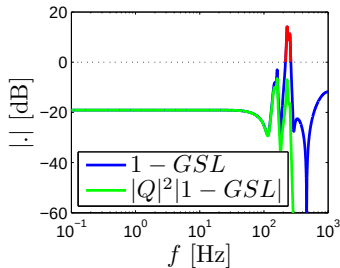
2. design Q such that

- for **performance** $Q(\omega) \approx 1$
- for **robustness**

$$|Q(1 - GSL)| < 1, \quad \forall \omega \in [0, 2\pi)$$

is satisfied

- also using non-parametric FRF $GS(\omega)$



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Key challenge: design L to approximate $(GS)^{-1}$

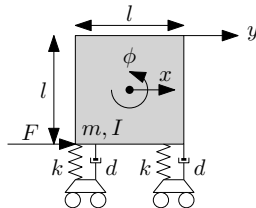
- ▶ what if GS nonminimum phase $\Rightarrow (GS)^{-1}$ unstable
- ▶ what if GS is strictly proper or has delays

The good news

- ▶ ILC can perfectly handle this: **noncausal** L
- ▶ this is why ILC can beat feedback: anticipate on disturbances!
- ▶ enabled by off-line computations

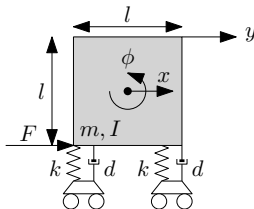
Benchmark example

- ▶ one NMP zero



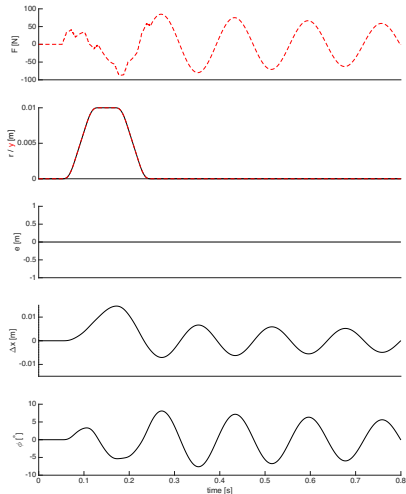
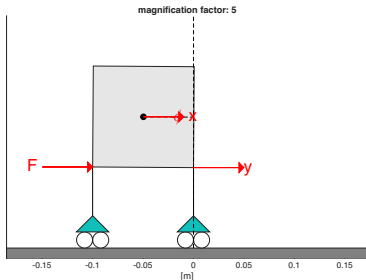
Simulation example

- ▶ flexible cart system
- ▶ varying sensor position
 - ▶ at bottom: collocated; **minimum** phase dynamics
 - ▶ at top: non-collocated; **non-minimum** phase dynamics
 - ▶ interpretation: depending on inertia/mass ratio, system either translates or rotates 'first' at sensor location
 - ▶ next: several movies illustrating consequences nonminimum phase dynamics
 - ▶ more details in (van Zundert & Oomen2017a)



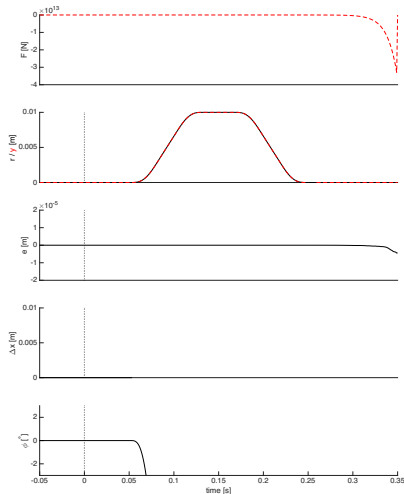
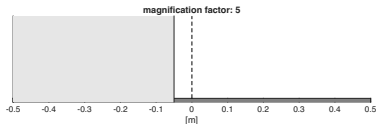
Collocated system

- ▶ collocated
 - ▶ minimum phase
- ▶ approach
 - ▶ regular inverse
- ▶ result
 - ▶ perfect tracking
 - ▶ bounded input
 - ▶ causal



Non-collocated system

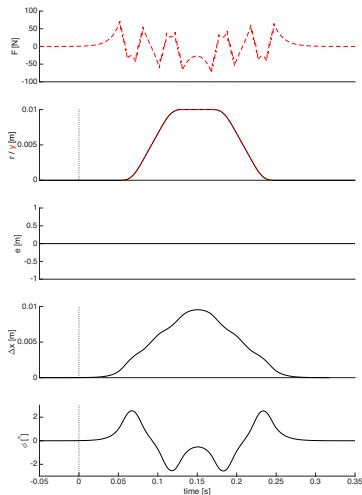
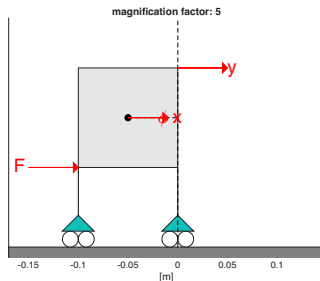
- ▶ non-collocated
 - ▶ **non**-minimum phase
- ▶ approach
 - ▶ regular inverse
- ▶ result
 - ▶ perfect tracking
 - ▶ **un**bounded input
 - ▶ causal



Non-collocated system

- ▶ non-collocated
 - ▶ **non**-minimum phase
- ▶ approach
 - ▶ stable inverse
- ▶ result
 - ▶ perfect tracking
 - ▶ bounded input
 - ▶ **non**causal

Key point: non-causality via L (and Q) (van Zundert & Oomen 2018)



Iterative Learning Control - What Can Be Achieved?

Traditional Motion Control

ILC: Basic Principles

Convergence Analysis

Enforcing Robustness

Implementation Aspects

Iteration-Variant Disturbances*

Alternative setup: serial ILC*

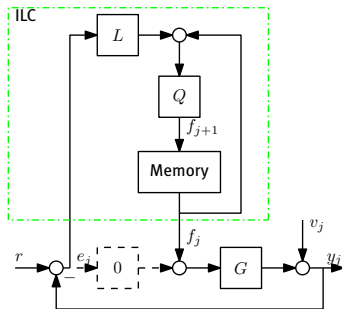
Multivariable ILC*

Summary

Disturbance properties

- ▶ tacit assumption: disturbances iteration **invariant**
- ▶ what if disturbances are iteration **variant** (Oomen & Rojas2017)

ILC scheme



Similar setup

- ▶ v_j : iteration **variant** disturbance
- ▶ r : iteration **invariant** 'disturbance'
- ▶ $G \in \mathcal{RH}_\infty$, $C = 0 \Rightarrow S = 1$
- ▶ $L, Q \in \mathcal{RH}_\infty$

ILC equations

- ▶ $e_j = Sr - Sv_j - GSf_j$
- ▶ $f_{j+1} = Q(f_j + Le_j)$
 $= Q((1 - LG)f_j + Lr - Lv_j)$

Why is this challenging to analyse?

- ▶ $f_{j+1} = Q((1 - LG)f_j + Lr - Lv_j)$
- ▶ initialization: $f_1 = 0$

Successive substitution

$$j=1. \quad f_2 = Q(Lr - Lv_1)$$

$$j=2. \quad f_3 = Q((1 - LG)f_2 + Lr - Lv_2) \\ = (Q(1 - LG) + 1)QLr - Q(1 - LG)QLv_1 - QLv_2$$

$$j. \quad f_{j+1} = \underbrace{\sum_{i=1}^j (Q(1 - LG))^{i-1} QLr}_{\text{straightforward part}} - \underbrace{\sum_{n=1}^j (Q(1 - LG))^{j-n} QLv_n}_{\text{all past noise realisations...!}}$$

Why is this challenging to analyse?

► at iteration j :
$$f_{j+1} = \underbrace{\sum_{i=1}^j (Q(1 - LG))^{i-1} QLr}_{\text{straightforward part}} - \underbrace{\sum_{n=1}^j (Q(1 - LG))^{j-n} QLv_n}_{\text{all past noise realisations...!}}$$

Result^(Oomen & Rojas2017)

- Assume v_j has iteration invariant properties
 - $v_j = Hn_j$, n_j i.i.d. ZMWN, variance λ_e
 - H monic, bistable
 - leads to spectrum $\phi_v = |H|^2 \lambda_e$
- Assume $|Q(1 - GL)| < 1$ (convergence)

Then, for $j \rightarrow \infty$,
$$\phi_{e_\infty} = \left| \frac{1-Q}{1-Q(1-LG)} \right|^2 \phi_r + \left(1 + \frac{|GQL|^2}{1-|Q(1-LG)|^2} \right) \phi_v$$

Result

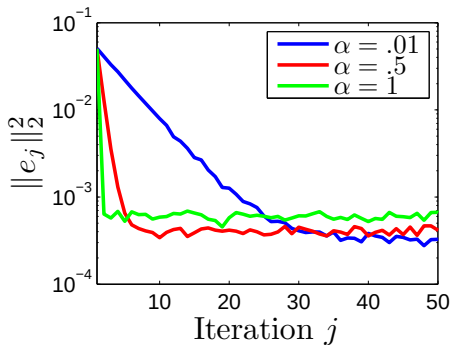
$$\phi_{e_\infty} = \left| \frac{1-Q}{1-Q(1-LG)} \right|^2 \phi_r + \left(1 + \frac{|GQL|^2}{1-|Q(1-LG)|^2} \right) \phi_v$$

Interpretation - 1

- ▶ assume $Q = 1, L = G^{-1}$
- ▶ $\phi_{e_\infty} = (1 + \frac{1}{1})\phi_v = 2\phi_v$
- ▶ learning **amplifies** iteration-variant disturbances!

Interpretation - 2

- ▶ assume $Q = 1, L = \alpha G^{-1}$
- ▶ $\phi_{e_\infty} = (1 + \frac{\alpha^2}{1-|1-\alpha|^2})\phi_v = (1 + \frac{\alpha^2}{-\alpha^2+2\alpha})\phi_v$
- ▶ for small α : $\phi_{e_\infty} \approx (1 + \frac{1}{2}\alpha)\phi_v$
- ▶ decreasing $\alpha \Rightarrow$ reducing asymptotic error



Reducing α

- ▶ reduced learning speed
- ▶ reduced amplification trial-variant disturbances

Example

- ▶ $f_{j+1} = f_j + Le_j$
- ▶ $L = \alpha(GS)^{-1}$
- ▶ $\alpha = 0.01, 0.5, 1$

Steady state error

α	$\frac{1}{N} \sum_{j=n+1}^{n+N} \ e_j\ _2^2$
0.01	$0.32 \cdot 10^{-3}$
0.5	$0.40 \cdot 10^{-3}$
1	$0.60 \cdot 10^{-3}$

Trade-off iteration-(in)variant disturbances for case $L = \alpha(GS)^{-1}$, $Q = 1$

- ▶ learning speed w.r.t. iteration invariant disturbances
- ▶ amplification of iteration variant disturbances

Extended ILC algorithms for iteration-variant disturbances

- ▶ high order ILC: $f_{j+1}(f_j, e_j, e_{j-1}, \dots)$
- ▶ stochastic approximation
 - ▶ $f_{j+1} = Q(f_j + \gamma_j L e_j)$
 - ▶ $\gamma_j \in \mathbb{R}_+$ decreasing sequence
- ▶ sparse ILC^(Oomen & Rojas2017)

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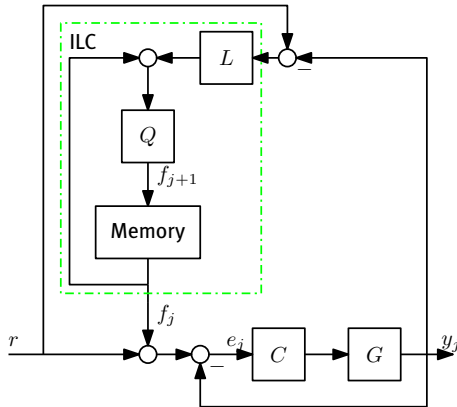
Implementation Aspects

Iteration-Variant Disturbances*

Alternative setup: serial ILC*

Multivariable ILC*

Summary



Alternative implementation

- ▶ serial instead of parallel ILC
- ▶ similar theory
- ▶ suitable for dedicated hardware with only access to reference signal
- ▶ sometimes advantageous if ILC measured signal is not equal to feedback measured signal

(Bolder & Oomen 2016)

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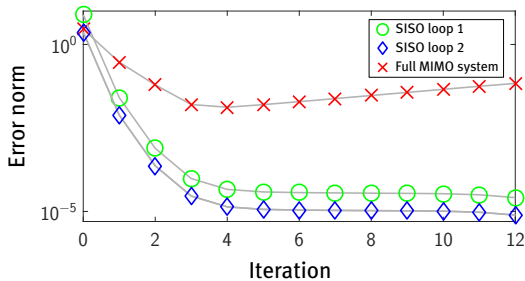
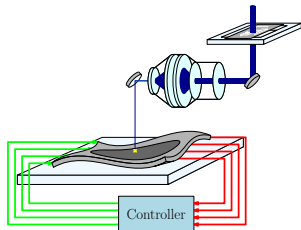
Multivariable ILC

- ▶ future mechatronic systems have many inputs and outputs
- ▶ tempting: use well-known SISO techniques to learn loop-by-loop



This does not work!

What's wrong?



What's wrong? interaction!

- much worse for ILC: 'bandwidth' up to Nyquist frequency

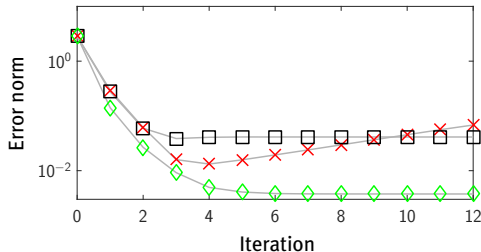
Recall: feedback design procedure^(Oomen2018a)

1. Traditional: ignore interaction
2. Static decoupling
3. Manual design for interaction
4. Optimal & robust control

Balances **modeling cost** vs. control performance

Multivariable ILC design ^(Blanken & Oomen2020)

	Performance	Modeling
SISO ignore	no 🙅	SISO 👍
Decentralized	good 😊	SISO 👍
Centralized	excellent 👍	MIMO 🙅



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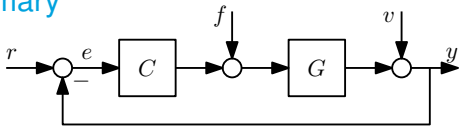
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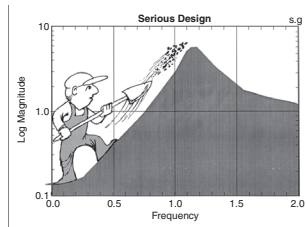
Multivariable ILC*

Summary

Summary



- ▶ $e = S(r - Gf) + Sv$
- ▶ feedback:
 - ▶ deal with **non**-repetitive disturbances
 - ▶ attenuate v : if $v = He$, e ZMWN, then $S \approx H^{-1}$ (Boeren et al.2017) (van Zundert & Oomen2017b)
- ▶ setpoint generator: select r , e.g., for point-to-point motion (Lambrechts et al.2005) (Boeren et al.2014)
- ▶ feedforward:
 - ▶ determine θ such that $f = F(\theta)r$, with $F(\theta) \approx G^{-1}$ (parameterized feedforward)
 - ▶ determine f such that $f \approx G^{-1}r$ via iteration-domain feedback (ILC) (all **repetitive** disturbances)



Summary

- ▶ basic ILC principle introduced
- ▶ convergence analysis:
 - ▶ model errors: need for robustness
 - ▶ design variables: L and Q
- ▶ iteration invariant vs. iteration variant disturbances
- ▶ developed framework
 - ▶ infinite time
 - ▶ L , Q : non-causal LTI filters
 - ▶ frequency domain/loopshaping design techniques

Reading

- ▶ ILC design procedure from these slides: (Oomen2020b) (Blanken et al.2019)
- ▶ general background on learning: (Oomen2018b)
- ▶ inversion techniques overview: (van Zundert & Oomen2018)
- ▶ noise analysis: (Oomen & Rojas2017)
- ▶ general ILC overview paper: (Bristow et al.2006)
- ▶ basics motion feedback and feedforward: (Oomen2020a)
- ▶ advanced motion control developments: (Oomen2018a)

Outlook

- ▶ finite time ILC analysis and synthesis
- ▶ L , Q : non-causal LTV filters
- ▶ and its extensions
 - ▶ LTV/position-dependent ILC (van Zundert, Bolder, Koekebakker & Oomen2016)
 - ▶ resource-efficient ILC (van Zundert, Bolder, Koekebakker & Oomen2016)
 - ▶ ILC for for varying tasks (van Zundert, Bolder & Oomen2016) (Bolder & Oomen2015)

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